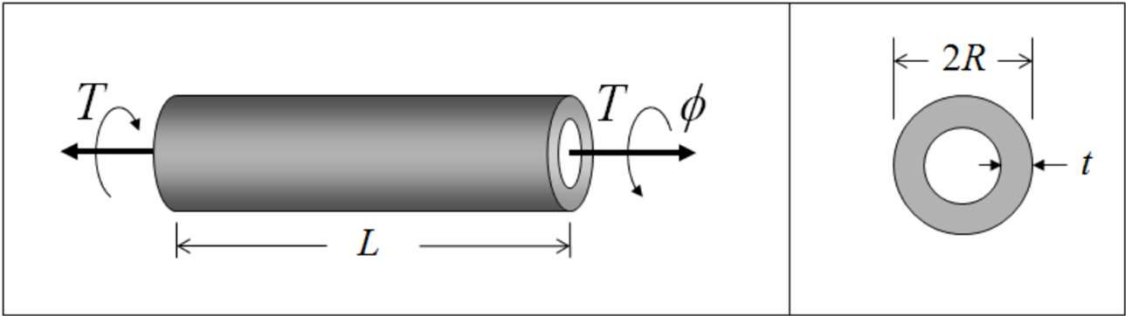


PART 1

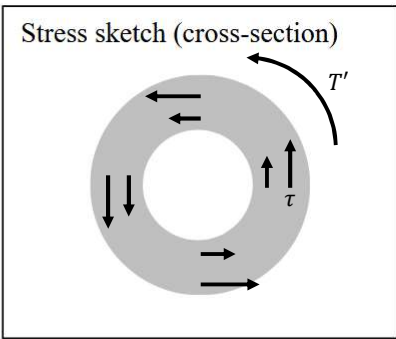
1. (20 pts). The sketch below shows the central section of an automotive anti-roll bar, which twists under car body roll, imparting a resisting torque during cornering. The anti-roll bar is a hollow shaft carrying torque T , resulting in a shaft twist ϕ . The shaft cross-section outer radius is R , wall thickness is t , and length is L .



1

1. What is the primary type of stress (shear stress, normal stress, or thermal stress?) that carries the anti-roll bar torque? Sketch the stress direction on a cross-section of the anti-roll bar.

Shear Stress



2. Give an equation that relates torque T , twist ϕ , and the shaft material properties and dimensions. Define any terms in the equation.

$$T = \frac{GJ\phi}{L}$$

Where

- G is the shear modulus
- J is the polar moment of inertia
- ϕ is the twist angle
- L is the length of the shaft

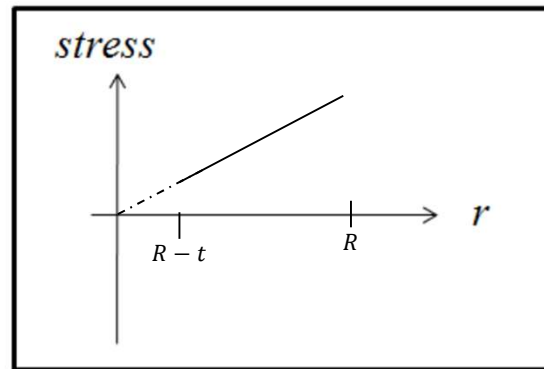
$$J = \frac{\pi}{2}(c_o^4 - c_i^4)$$

$$J = \frac{\pi}{2}(R^4 - (R - t)^4)$$

2

3. Sketch the distribution of the stress with radial position in the shaft cross-section.

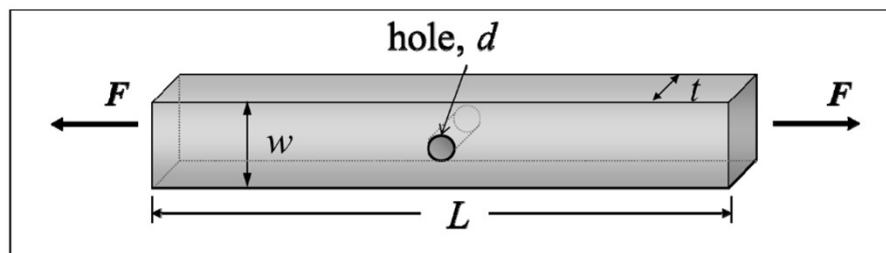
$$\tau = \frac{T\rho}{J}$$



3

2. (20 pts). Consider the prismatic member shown below, with a through hole of diameter d , subject to equal and opposite forces of magnitude F . (No external load is applied to the hole.) The member is designed such that the maximum allowable force, F_{all} , will not cause yielding of the material, with some factor of safety FS .

Assuming that all other parameters are held constant, assess the impact of an increase in any of the following properties or dimensions on **Factor of Safety, FS** . (Note: a variable effect means an increase in FS for some range of the variable, a decrease in another range. You must briefly explain any variable effects to get credit.) (Hint: it may be helpful to derive an equation for factor of safety FS , starting from the maximum allowable stress σ_{all} under load F_{all} , and the yield stress σ_Y , then substituting in the relevant design parameters.)



4

$$FS = \frac{Ultimate\ Load}{Allowable\ Load} = \frac{\sigma_Y}{\sigma_{all}}$$
$$FS = \frac{\sigma_Y}{K\sigma_{ave}} = \frac{\sigma_Y}{K\frac{F_{all}}{A_{min}}} = \frac{\sigma_Y A_{min}}{KF_{all}}$$

$$FS = \frac{\sigma_Y(w-d)t}{KF_{all}}$$

Note that E, G, ν, L, α don't have an effect on the internal stress on the member since it is only under the applied load and we don't have to consider deflections.

Since the stress concentration factor K decreases as the ratio of d/D increases, this implies that as the diameter increases, the stress concentration in the denominator will decrease, increasing FS . However, since the diameter is also in the numerator, as it increases, it will decrease FS . So, the effect of the diameter must be variable.

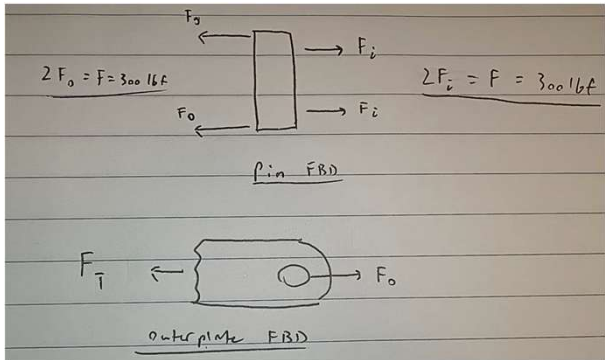
Note that Poisson's ratio can cause a slight decrease in the cross-sectional area, thus having a slight decrease in the factor of safety.

5

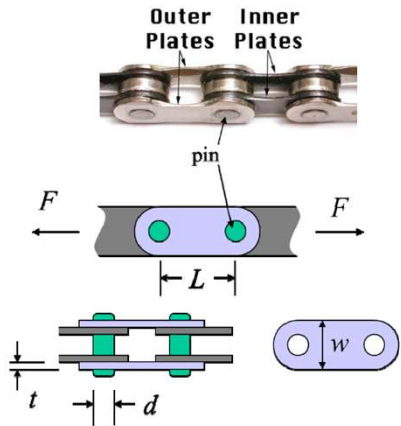
Increased Parameter Property, or Dimension:	Effect on Factor of Safety FS :			
	Increase	Decrease	No effect	Variable
Young's Modulus (E)			X	
Shear Modulus (G)			X	
Yield Stress (σ_Y)	X			
Poisson Ratio (ν)			X	
Length (L)			X	
Section Width (w)	X			
Section thickness (t)	X			
Thermal Expansion Coeff (α)			X	
Allowable Force (F_{all})		X		
Hole diameter (d)				X

6

PART 2 1. (30 points) A bicycle chain consists of pairs of inner and outer plates, each with a central hole, connected by cylindrical pins (see figure). Analyze the stress and factor of safety of the outer plates, which are assumed to be the weakest part of the chain. The chain is under total tension (carried by pairs of plates) $F = 300 \text{ lb}$. The outer plate dimensions are width $w = 0.315 \text{ in}$, plate pin-to-pin length $L = 0.5 \text{ in}$, and thickness $t = 0.0625 \text{ in}$, and the pin diameter is $d = 0.125 \text{ in}$. If the plates are made from steel with yield strength 65 ksi, estimate the maximum stress and calculate the factor of safety in the outer plate. The stress concentration factors for tensile stress and bearing stress from Amesweb are given in a chart for your convenience.



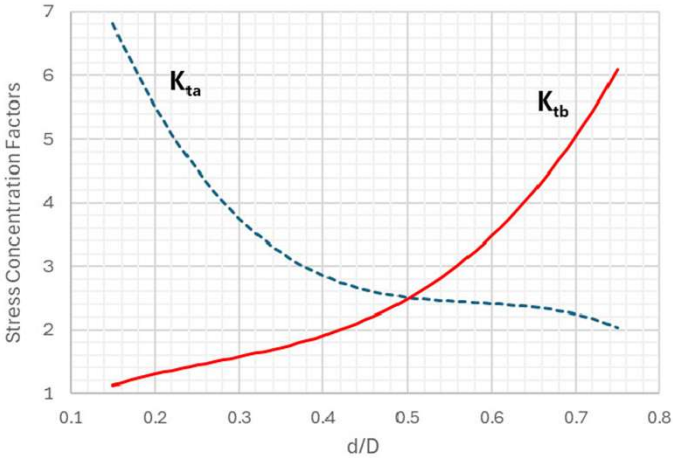
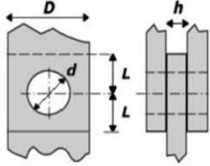
$$F_T = F_o = \frac{F}{2} = 150 \text{ lbf}$$



7

Plate with Closely Fitting Pin

$$\sigma_{bearing_{max}} = K_{tb} \sigma_{bearing}$$
$$\sigma_{tensile_{max}} = K_{ta} \sigma_{ave_{max}}$$



Based on <https://amesweb.info/stress-concentration-factor-calculator/pin-joint.aspx>

8

a. Calculate the maximum stress:

The outer plates are in tension, so the max average normal stress is

$$\sigma_{avg,max,norm} = \frac{F_T}{(w-d)t} = \frac{[150 \text{ lbf}]}{[0.315 - 0.125 \text{ in}][0.0625 \text{ in}]} = 12631.5789 \text{ psi}$$

We will also need to consider the bearing stress

$$\sigma_{avg,bearing} = \frac{F_T}{td} = \frac{[150 \text{ lbf}]}{[0.0625 \text{ in}][0.125 \text{ in}]} = 19200 \text{ psi}$$

Multiply by the stress concentration factor to find the max stresses

$$\frac{d}{D} = \frac{d}{w} = \frac{[0.125 \text{ in}]}{[0.315 \text{ in}]} = 0.3968 \quad K_{ta} = 2.95 \quad K_{tb} = 1.85$$

Also consider the shear stress inside the pin due to *double shear*

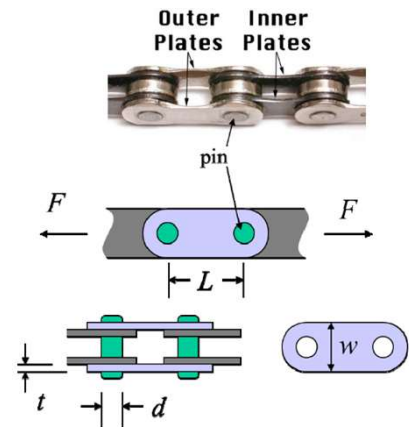
$$\tau_{ave,pin} = \frac{F/2}{\left(\frac{\pi}{4}d^2\right)} = \frac{F_T}{\left(\frac{\pi}{4}d^2\right)} = \frac{[150 \text{ lbf}]}{\frac{\pi}{4}[0.125 \text{ in}]^2} = 12223.0999 \text{ psi}$$

$$\sigma_{max,norm} = K_{ta}\sigma_{avg,max,norm} = (2.95)[12631.5789 \text{ psi}] = 37262.1577 \text{ psi}$$

$$\sigma_{max,bearing} = K_{tb}\sigma_{avg,bearing} = (1.85)[19200 \text{ psi}] = 35520 \text{ psi}$$

Max Stress is

$$\sigma_{max,norm} = 37262.1577 \text{ psi}$$



9

b. Calculate the factor of safety:

$$FS = \frac{\text{Ultimate Load}}{\text{Allowable Load}} = \frac{\sigma_Y}{\sigma_{max}}$$

$$FS = \frac{\left[65 \text{ ksi} \cdot \frac{10^3 \text{ psi}}{1 \text{ ksi}}\right]}{[37262.1577 \text{ psi}]} = 1.7443$$

c. Discuss your results: identify the critical location, comment on the accuracy of your estimate, comment on the quality of the design.

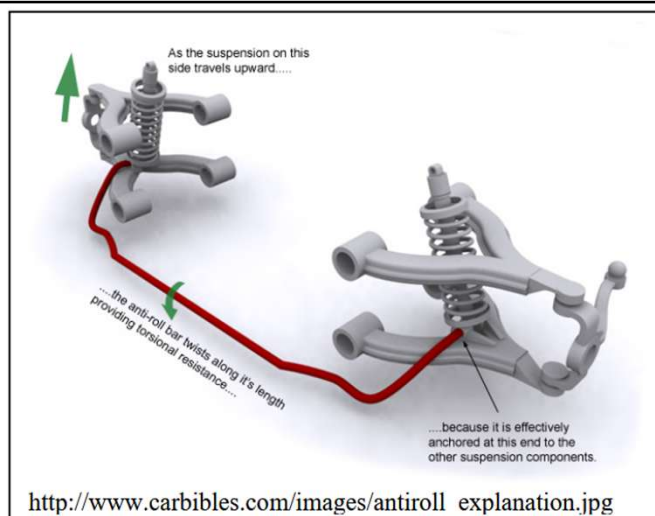
Considering all the stresses and stress concentrations, the critical location will be around the pin joint due to tensile stress. Since the max stress is less than the yield stress, the links are still in the elastic region, and the estimate of stress concentration is valid. Considering the factor of safety, this design is relatively robust under the current loading.

Design seems efficient because $\sigma_{max,bearing} \approx \sigma_{max,tensile}$. The calculation accuracy is somewhat limited because the pin isn't symmetrical on the plate and is instead closer to the ends. This is still a much better estimate than just using average stresses.

10

2. (30 pts.) The figure shows an automobile Sway Bar, also known as an anti-roll bar. The bar resists body roll during cornering, which improves handling. Whenever the car body rolls, the central part of the bar is twisted due to the body motion relative to the suspension.

- Given that the length of the bar is 1.42m, select the diameter of a round solid steel bar ($G = 77.2 \text{ GPa}$) such that when a 300 N-m torque is applied to the bar, the twist is 12° .
- Under these conditions, calculate the maximum shear stress in the bar.
- Is this design governed primarily by Stiffness, Strength, or Stability?



11

- a. Select diameter

Using the torque-twist relationship, I can find the diameter using the polar moment of inertia

$$T = \frac{GJ\phi}{L} \quad J = \frac{\pi}{2}(c^4) = \frac{\pi}{2}\left(\frac{d^4}{16}\right) = \frac{\pi}{32}d^4$$

$$T = \frac{G\phi}{L} \frac{\pi}{32} d^4 \longrightarrow d = \sqrt[4]{\frac{32LT}{\pi G\phi}}$$

$$d = \sqrt[4]{\frac{32[1.42 \text{ m}][300 \text{ N} \cdot \text{m}]}{\pi \left[77.2 \text{ GPa} \cdot \frac{10^9 \text{ Pa}}{1 \text{ GPa}} \right] \left[12^\circ \cdot \frac{\pi \text{ rad}}{180^\circ} \right]}} = 0.02276 \text{ m}$$

$$\boxed{d = 22.76 \text{ mm}}$$

- b. Max stress

The max shear stress is at the radius of the shaft and is given by

$$\tau = \frac{Tc}{J} = T \left(\frac{d}{2} \right) \frac{32}{\pi d^4} = \frac{16T}{\pi d^3} \quad \boxed{\tau = 129.5809 \text{ MPa}}$$

$$\tau = \frac{16[300 \text{ N} \cdot \text{m}]}{\pi[0.02276 \text{ m}]^3} = 129580908.8 \text{ Pa}$$

12

c. Design

The design is governed primarily by stiffness, since this shaft's purpose is to deflect and resist torques generated from the suspension, strength and stability aren't the primary roles. Thus, we are more interested in how much it deflects/twists under differing torques which is given by the torsional stiffness.

3. (10 pts extra credit) Refer to the bike chain in problem 1.

- a. Estimate the amount of stretch in one plate under the given load.

ASTM-A913 Grade 65 steel has an identical yield strength of 65 ksi so I will use its modulus of elasticity $E = 29 \times 10^6 \text{ psi}$ to estimate the amount of deflection. To find the total deflection in the outer plate, I will consider the deflection of the region between the pins since that's the majority of the material.

$$\delta = \frac{PL}{EA} = \frac{F_T L}{E(wt)} = \frac{[150 \text{ lbf}][0.5 \text{ in}]}{[29 \times 10^6 \text{ psi}][0.315 \text{ in}][0.0625 \text{ in}]} = 0.0001313 \text{ in} \quad \boxed{\delta = 0.1313 \times 10^{-3} \text{ in}}$$

- b. List one or more modifications to the design that could make the chain lighter without increasing the maximum stress, and discuss why the modification wouldn't increase the stress as well as how these would affect the chain stretch

Since the critical area/max stress is the tensile stress around the pinned joints, one way to decrease mass/volume is to make the middle region skinnier since the stress there is much lower. This won't affect the max stress at the pinned joints, but it'll decrease the center area which will increase the deflection of the plate.